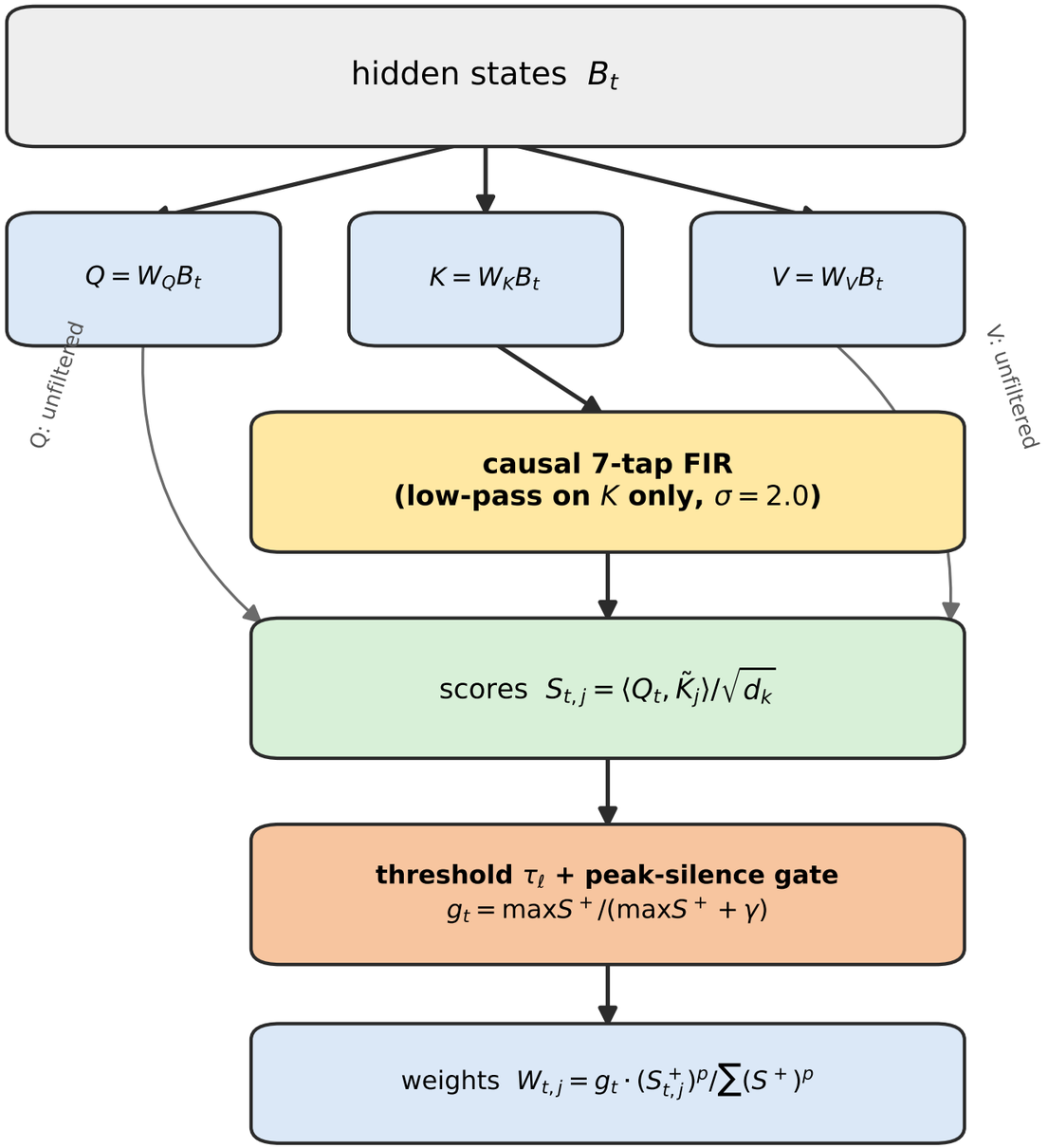
pn_fir_tau: a causal FIR on K, then a calibrated off switch

The filter acts on the key stream only; Q and V stay sharp.



What this buys you

One layer binds a queried key to the adjacent value slot.

Non-bijective recall: 44.5% → 100.0%

Rectification alone starves under the score shift: fixed $\tau = 0$ drops to 64.4%, with 91.3% negative scores and 44.5% dead rows.

One learned τ per layer restores 100% recall and keeps unanswerable rows silent: row mass 0.002 vs 0.100 for virtual sinks.

Schematic; not to scale. Numbers from the controlled experiments in the paper.